Md Farabi Mahbub

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Education

Bachelor of Science in Computer Science & Engineering, North South University May 2018 – Dec 2022

- CGPA: 3.81/4.00
- Specialization: Machine Learning, Deep Learning
- Honors: *summa cum laude*

Higher Secondary Certificate (HSC), Notre Dame College, Dhaka 2015 – 2017

- GPA: 5.00/5.00

Secondary School Certificate (SSC), Sirajganj Collectorate School & College 2015

- GPA: 5.00/5.00

Teaching & Research Experience

Research Assistant, Department of Electrical & Computer Engineering, North South University April 2026 –
Supervisor: Dr. Ahsanur Rahman
Research Area: Geospatial Machine Learning, Remote Sensing

- Developing AI systems for groundwater contamination risk assessment in Bangladesh
- Integrating satellite Earth observation, terrain, and hydrological modeling, and national geological survey data to predict contamination from remotely sensed signals alone.
- Developing a public web application to make risk screening accessible to communities and field workers without laboratory access.

Trainee Research Associate (Young Researchers' Fellowship), Data and Innovation Research Program, Brac Institute of Governance and Development Dec 2025 – March 2026

- Received formal training in statistics, advanced econometrics, and research methodology, including end-to-end survey research design and data analysis workflows.
- Developed machine learning and deep learning solutions, including a multi-model ensemble pipeline for detecting waste fields from multi-source satellite imagery.

Lab Instructor, Department of Electrical & Computer Engineering, North South University Feb 2024 – Jan 2026

- Instructed undergraduate lab courses in C Programming, Data Structures & Algorithms, and Database Management Systems, and developed associated lab manuals and instructional materials.
- Designed, administered, and graded quizzes and mid-term and final examinations.
- Supervised and evaluated student projects.

Research Assistant, Department of Electrical & Computer Engineering, North South University June 2024 – Dec 2024
Supervisor: Dr. Ahsanur Rahman
Research Area: Machine Learning, Generative Models

- Applied machine learning techniques to analyze environmental data and assess groundwater contamination risk based on geospatial and geological factors.
- Explored generative models to improve model performance under data scarcity.

Teaching Assistant, Department of Electrical & Computer Engineering, North South University April 2021 – Sep 2022

- Conducted tutorial sessions for students requiring additional help.

- Assisted faculty members in grading assignments, setting assessments, performing proctoring duties, and handling course-related administrative tasks.

Publications

Md. Farabi Mahbub, Warsi Omrao Khan Shuvo, and Sifat Momen, "Prediction of Glycemic Control in Diabetes Mellitus Patients Using Machine Learning," *Data Analytics in System Engineering: Proceedings of 7th Computational Methods in Systems and Software 2023*, Vol. 4, pp. 216, Springer, Cham, 2024.

Technical Skills

Languages: C, C++, Java, Python, Assembly

ML Libraries: Scikit-learn, Pytorch

Statistical Tools: Stata

Data Analysis & Visualization: Pandas, Seaborn, Matplotlib

Web Frameworks: Flask, Django

Database: MySQL, PostgreSQL

Documentation: LaTeX, MS Word, PowerPoint, Excel

Language Proficiency

IELTS Academic: Overall Band Score 8.0 (Listening 8.5, Reading 8.0, Writing 7.5, Speaking 7.5) October, 2025

Selected Projects

From Data to Safety: A Machine Learning Framework for Groundwater Arsenic Prediction: We developed a machine learning framework to predict arsenic contamination in groundwater using pre-digging parameters, eliminating the need for costly water sampling, addressed data scarcity with CGAN-based synthetic data augmentation, incorporated Explainable AI for interpretability and deployed the best-performing model. Technology: Scikit-learn, GAN, Django.

Deep Learning-Based Identification and Quantification of Visual Pollution: The goal of this project was to develop a deep learning-based system for detecting and quantifying visual pollution in urban environments. We compiled a labeled dataset from Google Street View and augmented it for model training. Multiple YOLO-based object detection models were trained to identify pollutants. A prototype software application was built to integrate YOLOv7 for real-time object detection from video footage captured by moving vehicles. The system tracks and counts visually polluted objects, providing an interactive dashboard to monitor pollution. Technology: PyTorch, YOLOv7, Flask, Makesense.ai.

FridayHaat: An Interactive eCommerce Web Application: An eCommerce platform built on Django, with a custom user model, email activation, Google sign-in through allauth, and guest checkout for buyers who want no account. Stripe handles customers, cards and charges. Product search runs across title, description, category, price, manufacturer and tags in a single query. The cart, the contact form and the payment step call a JSON endpoint through jQuery, so those actions leave the page in place.

Deep Learning-Based Waste Field Detection Using Multi-Source Satellite Imagery: Developed a deep learning framework for detecting waste disposal sites from satellite imagery using multi-source remote sensing data. Satellite composites were generated using Sentinel-1 SAR and Sentinel-2 optical imagery, forming multi-channel inputs for model training. Implemented cross-validation, focal loss, and ensemble-based classification to address class imbalance and improve detection performance. Technology: PyTorch, Google Earth Engine, Sentinel-1/2, Raster Processing

Prediction of Glycemic Control in Diabetes Mellitus Patients using Machine Learning Techniques: This study focused on predicting glycemic control in diabetes patients using machine learning models. We used SMOTE to address data imbalance and incorporated explainable AI to gain insights into the decision-making process.

Technology: Scikit-learn, LIME, SHAP, Shapash.

Coin U-Net: Context Invariance Image Segmentation Using Deep Learning: This study aimed to improve biomedical image segmentation by addressing the limitations of traditional U-Net architectures. The Coin U-Net model introduces a context-guided approach using a Siamese network, training on medical images paired with masked versions to help the model distinguish between infected regions and healthy tissue while capturing both short- and long-range context. Technology: PyTorch, Python.

Awards

- Young Researchers' Fellowship, Brac Institute of Governance and Development 2025
- Graduated *summa cum laude*, North South University 2022
- Awarded 50% merit-based scholarship for admission test performance, North South University 2018
- Sonali Bank CEO & Managing Director Medal 2017
- Khwaja Mozammel Hoque (R) Foundation Scholarship 2014, 2012
- Bangladesh Government Talent Pool Scholarship (PSC Exam) 2009

References

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